



UCD Michael Smurfit Graduate Business School

MSc Business Analytics

Programming for Analytics MIS41110

Ukulele Tuesday

"Declaration of own work"

We declare that this material, which we now submit for assessment, is entirely our work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of our work.

Submission Date: 22 November 2024

Table of Contents

ABOUT THE UKULELE TUESDAY	3
Overview	3
TECHNICAL DOCUMENTATION	3
User Input	3
Error Handling	3
Variables and Naming Conventions	3
Key functionalities	4
Dataframe Results	4
GUI – USER MANUAL	5
UML ACTIVITY DIAGRAM	6
REFERENCES	7

1. ABOUT UKULELE TUESDAY

1.1 Overview

The Ukulele Tuesday Data Explorer is an interactive application designed to explore, analyze, and visualize song data. It allows users to load multiple datasets, filter data based on date ranges, and generate meaningful visualizations.

1.2 Purpose

This project aims to provide an intuitive and dynamic platform for analyzing song metadata, historical play data, and user requests. The application supports decision-making and data-driven insights for music analytics.

2. TECHNICAL DOCUMENTATION

2.1 User Input

The application processes three datasets:

- `tabddb.csv`: Contains song metadata, including artist, genre, difficulty, and language.
- `playdb.csv`: Tracks the history of song plays with timestamps.
- `requestdb.csv`: Records user requests for songs during specific sessions.

Users can load these datasets through the 'Load Data' buttons in the GUI.

2.2 Error Handling

The application includes robust error-handling mechanisms to ensure a smooth user experience.

Examples of error handling:

- Invalid file format or missing columns in the CSV files trigger a detailed error message.
- If no data matches the selected filters, a notification is displayed to guide the user.
- Improper date selection is restricted by disabling out-of-range values.

2.3 Variables and Naming conventions

Variables in the program are named descriptively for clarity and maintainability.

Key naming conventions include:

- ``df_``: Data Frames for handling datasets.
- ``fig_``: Figure objects for Matplotlib visualizations.
- ``filtered_``: Prefix for filtered datasets.

This consistency ensures readability and reduces debugging time.

2.4 Key Functionalities

The application provides the following core functionalities:

- Data Loading: Users can load datasets dynamically.
- Date Filtering: Interactive calendar-based filters to specify the range of analysis.
- Visualization: Dynamic graphs for insights like difficulty distribution, duration trends, and play counts.
- Data Export: Save filtered data as a CSV file for further processing.

3. DATAFRAME RESULTS

The application processes the datasets and generates a consolidated DataFrame. The final DataFrame includes:

- Song metadata (e.g., artist, genre, difficulty).
- Play history with timestamps.
- User request data for specific sessions.

This DataFrame forms the basis for visualizations and is displayed in the GUI table.

GUI USER MANUAL:

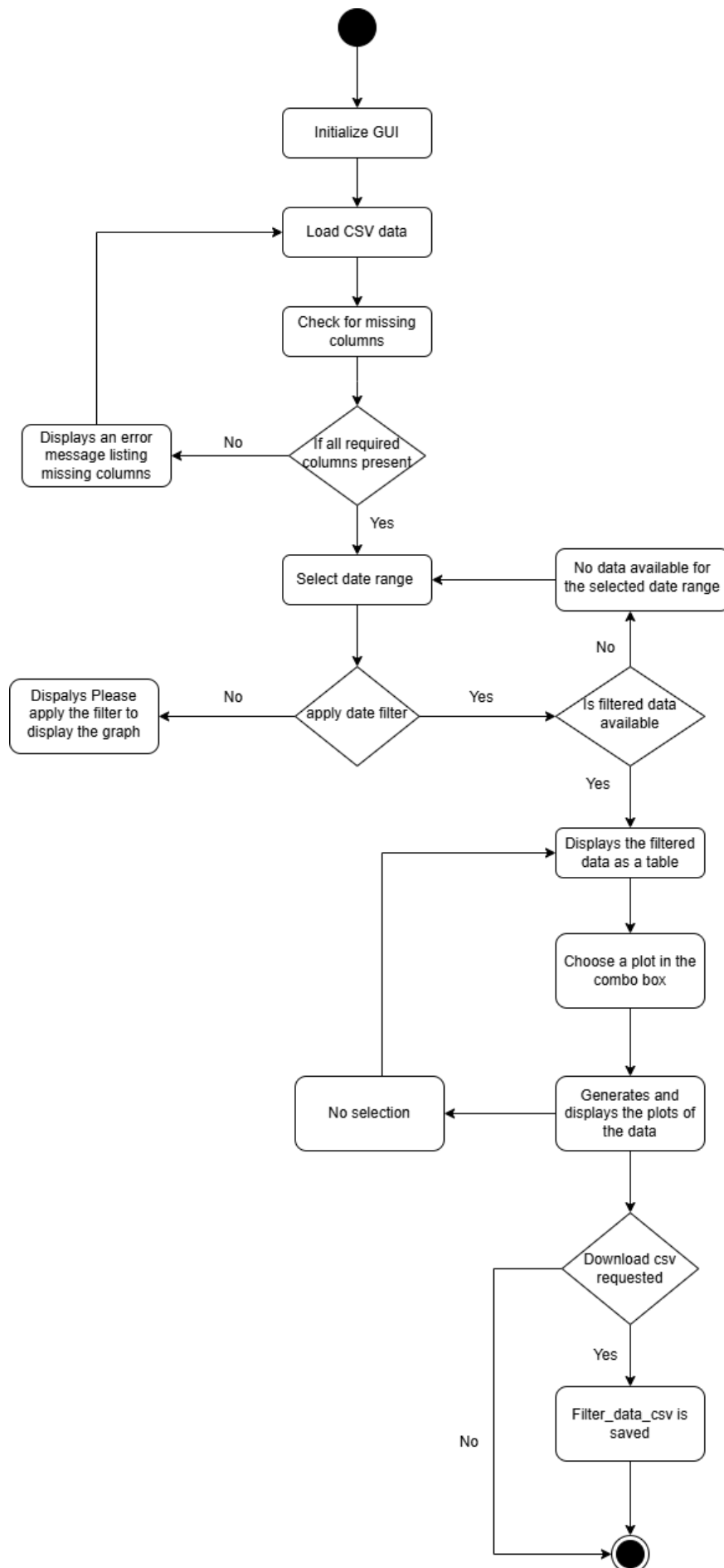
1. Install Python 3 (ensure the path is configured and download PIP).
2. Install the following modules/libraries in command prompt by following the Example: `py -m pip install matplotlib`

The required libraries are:

- numpy
- pandas
- plotly
- seaborn
- tkcalendar
- matplotlib
- Pillow

3. Run the program on Terminal, Powershell, or a Python IDE by running: `python main.py`
4. The popup screen appears welcoming you to the Ukulele Tuesday Data Explorer.
5. The interface provides buttons to load three datasets:
 - tabdb.csv: Song metadata.
 - playdb.csv: Song play history.
 - requestdb.csv: Song requests.
 - click each button to select and load the corresponding file. Error messages are displayed if required columns are missing or if files are not loaded correctly.
6. After loading playdb.csv, date filter will populate with the starting and ending date in data set as default.
 - Select the start and end dates to narrow down the data.
7. Then click the Apply Filter button to filter data based on the selected date range. This prepares the data for visualization and tabular display.
8. The filtered data is displayed in a table below the visualization area. The application will function smoothly, transitioning between actions like filtering, graph selection, and data export without displaying any error messages.
9. Below the date filter, there is a dropdown menu to select the type of graph you want to display (e.g., Difficulty level, Duration, Language, etc.).
10. After applying the filter, the selected graph will be displayed in the main visualization panel. Select the 'no selection' option from the dropdown to return to your table view.
11. Click the Download CSV button to save the filtered data as `filtered_data.csv` in the program directory.

UML Activity Diagram:



REFERENCES:

- Plotting library for bars and charts. Available at:
<https://matplotlib.org/>
- For donut chart. Available at:
<https://plotly.com/python/plotly-express/>
- Tkinter GUI - tools and integration. Available at:
<https://docs.python.org/3/library/tkinter.html>
- For debugging and error handling – ChatGPT 4o